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Stepwise Regression

Which variables should you drop from your model? This is a common question for all modeling, but especially logistic regression. In this section we will cover a popular variable selection technique - stepwise regression. This isn't the only possible technique, but will be the primary focus here.

We will be going back to the Ames, Iowa dataset for exploring these techniques.

Stepwise regression techniques involve the three common methods:

1. Forward Selection
2. Backward Selection
3. Stepwise Selection

These techniques add or remove (depending on the technique) one variable at a time from your regression model to try and "improve" the model. There are a variety of different selection criteria to use to add or remove variables from a logistic regression. Two common approaches are to use either p-values or one of the pair of AIC/BIC.

Although p-values are falling out of popularity, it is primarily because people often use the 0.05 significance level without any regards to sample size. Although 0.05 is a good significance level for a sample size around 50, this level should be adjusted based on sample size.

However, it can be shown that mathematically the AIC/BIC criterion for adding or removing variables with stepwise selection (which is becoming very popular) is the same thing as using p-values in likelihood ratio tests. AIC is calculated as follows:

$$AIC = -2\log(L) + 2p$$

where L is the likelihood function and p is the number of variables being estimated in the model. Let's compare two models - one with p variables and one with $p+1$ variables. Assuming the additional variable lowers AIC, we can see the following relationship:

$$AIC_{p+1} < AIC_p$$
$$-2\log(L_{p+1}) + 2(p+1) < -2\log(L_p) + 2p$$
$$2 < 2(\log(L_{p+1}) - \log(L_p))$$

The right hand side of the equation is a Likelihood Ratio Test that follows a χ^2_1 distribution. So we know the significance level from this LRT is the following:

$$1 - P(\chi^2_1 > 2) = 0.1573$$

This means that the AIC selection method for stepwise selection is mathematically the same as a p-value stepwise selection technique with a significance level (α level) of 0.1573 for continuous variables or binary categorical variables. More than two categories in a categorical variable would be greater than 1 degree of freedom in the calculation above.

The same math can be applied to the BIC selection technique as well. The BIC calculation is the following:

$$BIC = -2\log(L) + p \times \log(n)$$

Working through the math, this is also a Likelihood Ratio Test that follows a χ^2_1 distribution. So we know the significance level from this LRT is the following:

$$1 - P(\chi^2_1 > \log(n)) = \ldots$$

Notice how the significance level changes depending on sample size due to the BIC equation, which is exactly what is recommended for any selection technique using p-values.

With that all understood, let's look at the three common stepwise regression approaches.

Stepwise Selection

Here we will work through stepwise selection. In stepwise selection, the initial model is empty (contains no variables, only the intercept). Each variable is then tried to see if it is significant based on AIC/BIC or p-value with a specified significance level. The most significant variable (or the one that reduces the AIC/BIC the most) is then added to the model. Next, all variables in the model (here only one) are tested to see if they are still significant (or don't hinder the AIC/BIC if dropped). If not, they are dropped. If so, then the remaining variables are again tested and the next most impactful variable (p-value or AIC/BIC depending on the approach) is added to the model. This process repeats until either no more significant variables are available to add or the same variable keeps being added and then removed from the model based on AIC/BIC or p-value (depending on the approach).

Let's look at this approach using all of our softwares!

R Python SAS

R's `step` function traditionally uses either AIC or BIC for adding or removing variables from the stepwise regression techniques.

In R we must specify two models - the empty and full models - to step between. Notice the full model contains all the variables, while the empty model contains only the intercept (signified by `~ 1` in the formula). We then use the `step` function where we start at the empty model. The `scope` option tells R the lower (`lower =`) and upper (`upper =`) model to step between. The `direction = "both"` option tells R to use stepwise selection.

▼ Code

```
full.model <- glm(Bonus ~ Gr_Liv_Area + factor(House_Style) + G
                          Fireplaces + factor(Full_Bath) + fact
                          factor(Central_Air) + Second_Flr_SF +
                  data = train, family = binomial(link = "logit

empty.model <- glm(Bonus ~ 1, data = train, family = binomial(l

step.model <- step(empty.model,
                   scope = list(lower=formula(empty.model),
                                upper=formula(full.model)),
                   direction = "both")
```

```
Start:  AIC=2777.81
Bonus ~ 1

                      Df Deviance    AIC
+ factor(Full_Bath)    4   1911.5 1921.5
+ Gr_Liv_Area          1   1926.4 1930.4
+ Garage_Area          1   2135.4 2139.4
+ First_Flr_SF         1   2294.1 2298.1
+ Fireplaces           1   2423.7 2427.7
+ TotRms_AbvGrd        1   2449.7 2453.7
+ factor(House_Style)  7   2542.1 2558.1
+ factor(Half_Bath)    2   2608.1 2614.1
+ Lot_Area             1   2621.9 2625.9
+ Second_Flr_SF        1   2631.8 2635.8
+ factor(Central_Air)  1   2654.3 2658.3
<none>                     2775.8 2777.8

Step:  AIC=1921.48
Bonus ~ factor(Full_Bath)

                      Df Deviance    AIC
+ Garage_Area          1   1616.3 1628.3
+ Fireplaces           1   1659.5 1671.5
+ Gr_Liv_Area          1   1665.7 1677.7
+ First_Flr_SF         1   1711.7 1723.7
+ Lot_Area             1   1811.5 1823.5
+ factor(Half_Bath)    2   1812.2 1826.2
+ factor(Central_Air)  1   1827.6 1839.6
+ factor(House_Style)  7   1822.3 1846.3
+ TotRms_AbvGrd        1   1885.6 1897.6
+ Second_Flr_SF        1   1908.2 1920.2
<none>                     1911.5 1921.5
- factor(Full_Bath)    4   2775.8 2777.8

Step:  AIC=1628.26
Bonus ~ factor(Full_Bath) + Garage_Area

                      Df Deviance    AIC
+ Fireplaces           1   1440.9 1454.9
+ Gr_Liv_Area          1   1481.3 1495.3
+ First_Flr_SF         1   1540.6 1554.6
+ factor(Half_Bath)    2   1546.5 1562.5
+ factor(House_Style)  7   1571.5 1597.5
+ factor(Central_Air)  1   1585.6 1599.6
+ Lot_Area             1   1589.0 1603.0
+ TotRms_AbvGrd        1   1606.6 1620.6
+ Second_Flr_SF        1   1608.7 1622.7
<none>                     1616.3 1628.3
- Garage_Area          1   1911.5 1921.5
- factor(Full_Bath)    4   2135.4 2139.4

Step:  AIC=1454.86
Bonus ~ factor(Full_Bath) + Garage_Area + Fireplaces

                      Df Deviance    AIC
+ Gr_Liv_Area          1   1381.5 1397.5
+ factor(Half_Bath)    2   1389.8 1407.8
+ factor(House_Style)  7   1386.2 1414.2
+ First_Flr_SF         1   1413.8 1429.8
+ factor(Central_Air)  1   1426.2 1442.2
+ Lot_Area             1   1435.1 1451.1
+ Second_Flr_SF        1   1436.5 1452.5
<none>                     1440.9 1454.9
+ TotRms_AbvGrd        1   1440.3 1456.3
- Fireplaces           1   1616.3 1628.3
- Garage_Area          1   1659.5 1671.5
- factor(Full_Bath)    4   1943.5 1949.5

Step:  AIC=1397.55
Bonus ~ factor(Full_Bath) + Garage_Area + Fireplaces + Gr_Liv_Ar

                      Df Deviance    AIC
+ factor(House_Style)  7   1310.0 1340.0
+ TotRms_AbvGrd        1   1335.1 1353.1
+ factor(Central_Air)  1   1354.3 1372.3
+ factor(Half_Bath)    2   1356.4 1376.4
+ First_Flr_SF         1   1368.2 1386.2
+ Second_Flr_SF        1   1370.7 1388.7
<none>                     1381.5 1397.5
+ Lot_Area             1   1381.2 1399.2
- Gr_Liv_Area          1   1440.9 1454.9
- Fireplaces           1   1481.3 1495.3
- Garage_Area          1   1546.1 1560.1
- factor(Full_Bath)    4   1626.6 1634.6

Step:  AIC=1339.99
Bonus ~ factor(Full_Bath) + Garage_Area + Fireplaces + Gr_Liv_Ar
    factor(House_Style)

                      Df Deviance    AIC
+ factor(Half_Bath)    2   1261.2 1295.2
+ TotRms_AbvGrd        1   1268.8 1300.8
+ factor(Central_Air)  1   1288.2 1320.2
<none>                     1310.0 1340.0
+ Lot_Area             1   1309.7 1341.7
+ Second_Flr_SF        1   1309.9 1341.9
+ First_Flr_SF         1   1309.9 1341.9
- factor(House_Style)  7   1381.5 1397.5
- Gr_Liv_Area          1   1386.2 1414.2
- Fireplaces           1   1394.8 1422.8
- Garage_Area          1   1403.9 1431.9
- factor(Full_Bath)    4   1511.2 1533.2

Step:  AIC=1295.15
Bonus ~ factor(Full_Bath) + Garage_Area + Fireplaces + Gr_Liv_Ar
    factor(House_Style) + factor(Half_Bath)

                      Df Deviance    AIC
+ TotRms_AbvGrd        1   1228.3 1264.3
+ factor(Central_Air)  1   1248.4 1284.4
+ First_Flr_SF         1   1259.0 1295.0
<none>                     1261.2 1295.2
+ Second_Flr_SF        1   1260.3 1296.3
+ Lot_Area             1   1260.7 1296.7
- factor(Half_Bath)    2   1310.0 1340.0
- Gr_Liv_Area          1   1331.8 1363.8
- Fireplaces           1   1335.7 1367.7
- Garage_Area          1   1336.7 1368.7
- factor(House_Style)  7   1356.4 1376.4
- factor(Full_Bath)    4   1490.6 1516.6

Step:  AIC=1264.3
Bonus ~ factor(Full_Bath) + Garage_Area + Fireplaces + Gr_Liv_Ar
    factor(House_Style) + factor(Half_Bath) + TotRms_AbvGrd

                      Df Deviance    AIC
+ factor(Central_Air)  1   1218.9 1256.9
<none>                     1228.3 1264.3
+ First_Flr_SF         1   1226.7 1264.7
+ Lot_Area             1   1227.3 1265.3
+ Second_Flr_SF        1   1227.7 1265.7
- TotRms_AbvGrd        1   1261.2 1295.2
- factor(Half_Bath)    2   1268.8 1300.8
- Fireplaces           1   1291.2 1325.2
- Garage_Area          1   1294.7 1328.7
- factor(House_Style)  7   1315.8 1337.8
- Gr_Liv_Area          1   1330.3 1364.3
- factor(Full_Bath)    4   1453.8 1481.8

Step:  AIC=1256.88
Bonus ~ factor(Full_Bath) + Garage_Area + Fireplaces + Gr_Liv_Ar
    factor(House_Style) + factor(Half_Bath) + TotRms_AbvGrd +
    factor(Central_Air)

                      Df Deviance    AIC
<none>                     1218.9 1256.9
+ First_Flr_SF         1   1217.7 1257.7
+ Lot_Area             1   1218.0 1258.0
+ Second_Flr_SF        1   1218.5 1258.5
- factor(Central_Air)  1   1228.3 1264.3
- TotRms_AbvGrd        1   1248.4 1284.4
- factor(Half_Bath)    2   1253.1 1287.1
- Fireplaces           1   1272.7 1308.7
- Garage_Area          1   1272.9 1308.9
- factor(House_Style)  7   1300.0 1324.0
- Gr_Liv_Area          1   1324.1 1360.1
- factor(Full_Bath)    4   1428.7 1458.7
```

▼ Code

```
summary(step.model)
```

```
Call:
glm(formula = Bonus ~ factor(Full_Bath) + Garage_Area + Fireplac
    Gr_Liv_Area + factor(House_Style) + factor(Half_Bath) + TotR
    factor(Central_Air), family = binomial(link = "logit"), data

Coefficients:
                                  Estimate Std. Error z valu
(Intercept)                     -1.041e+01  1.537e+00  -6.77
factor(Full_Bath)1              -6.860e-01  1.325e+00  -0.51
factor(Full_Bath)2               1.894e+00  1.341e+00   1.41
factor(Full_Bath)3               4.152e+00  1.610e+00   2.57
factor(Full_Bath)4              -1.261e+00  2.493e+00  -0.50
Garage_Area                      3.583e-03  5.187e-04   6.90
Fireplaces                       9.142e-01  1.272e-01   7.18
Gr_Liv_Area                      3.827e-03  4.033e-04   9.48
factor(House_Style)One_and_Half_Unf -8.941e+00  3.682e+02  -0.02
factor(House_Style)One_Story     2.396e+00  3.285e-01   7.29
factor(House_Style)SFoyer        1.760e+00  6.382e-01   2.75
factor(House_Style)SLvl          1.105e+00  4.530e-01   2.43
factor(House_Style)Two_and_Half_Fin -4.855e-01  6.945e+00  -0.07
factor(House_Style)Two_and_Half_Unf  8.329e-01  8.891e-01   0.93
factor(House_Style)Two_Story     9.801e-01  3.380e-01   2.90
factor(Half_Bath)1               1.195e+00  2.153e-01   5.55
factor(Half_Bath)2              -1.301e-01  8.053e-01  -0.16
TotRms_AbvGrd                   -4.322e-01  8.128e-02  -5.31
factor(Central_Air)Y             1.620e+00  5.866e-01   2.76
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2775.8  on 2050  degrees of freedom
Residual deviance: 1218.9  on 2032  degrees of freedom
AIC: 1256.9

Number of Fisher Scoring iterations: 14
```

The model building summary is shown to have added variables based on lowest AIC. This may be different across different stepwise techniques.

Backward Selection

Here we will work through backward selection. In backward selection, the initial model is full (contains all variables, including the intercept). Each variable is then tried to see if it is significant based on AIC/BIC or p-value with a specified significance level. The least significant variable (or the one that hinders the AIC/BIC the most) is then dropped from the model. Then the remaining variables are again tested and the next least significant variable is dropped from the model. This process repeats until no more insignificant variables are available to drop or the AIC/BIC no longer improves with the deletion of another variable.

Let's look at this approach using all of our softwares!

R Python SAS

In R we must specify one model - the full model - to step backward from. Notice the full model contains all the variables. We then use the `step` function where we start at the full model. The `direction = "backward"` option tells R to use backward selection.

▼ Code

```
full.model <- glm(Bonus ~ Gr_Liv_Area + factor(House_Style) + G
                          Fireplaces + factor(Full_Bath) + fact
                          factor(Central_Air) + Second_Flr_SF +
                  data = train, family = binomial(link = "logit

back.model <- step(full.model, direction = "backward")
```

```
Start:  AIC=1257.6
Bonus ~ Gr_Liv_Area + factor(House_Style) + Garage_Area + Firepl
    factor(Full_Bath) + factor(Half_Bath) + Lot_Area + factor(Ce
    Second_Flr_SF + TotRms_AbvGrd + First_Flr_SF

                      Df Deviance    AIC
- Lot_Area             1   1214.2 1256.2
- Gr_Liv_Area          1   1214.3 1256.3
<none>                     1213.6 1257.6
- Second_Flr_SF        1   1217.1 1259.1
- First_Flr_SF         1   1217.8 1259.8
- factor(Central_Air)  1   1222.3 1264.3
- factor(House_Style)  7   1241.0 1271.0
- TotRms_AbvGrd        1   1243.0 1285.0
- factor(Half_Bath)    2   1248.1 1288.1
- Garage_Area          1   1262.0 1304.0
- Fireplaces           1   1262.7 1304.7
- factor(Full_Bath)    4   1425.9 1461.9

Step:  AIC=1256.18
Bonus ~ Gr_Liv_Area + factor(House_Style) + Garage_Area + Firepl
    factor(Full_Bath) + factor(Half_Bath) + factor(Central_Air)
    Second_Flr_SF + TotRms_AbvGrd + First_Flr_SF

                      Df Deviance    AIC
- Gr_Liv_Area          1   1214.9 1254.9
<none>                     1214.2 1256.2
- Second_Flr_SF        1   1217.7 1257.7
- First_Flr_SF         1   1218.5 1258.5
- factor(Central_Air)  1   1223.0 1263.0
- factor(House_Style)  7   1241.1 1269.1
- TotRms_AbvGrd        1   1243.2 1283.2
- factor(Half_Bath)    2   1248.7 1286.7
- Garage_Area          1   1264.2 1304.2
- Fireplaces           1   1265.2 1305.2
- factor(Full_Bath)    4   1425.9 1459.9

Step:  AIC=1254.91
Bonus ~ factor(House_Style) + Garage_Area + Fireplaces + factor(
    factor(Half_Bath) + factor(Central_Air) + Second_Flr_SF +
    TotRms_AbvGrd + First_Flr_SF

                      Df Deviance    AIC
<none>                     1214.9 1254.9
- factor(Central_Air)  1   1223.8 1261.8
- factor(House_Style)  7   1243.0 1269.0
- TotRms_AbvGrd        1   1244.7 1282.7
- Second_Flr_SF        1   1245.5 1283.5
- factor(Half_Bath)    2   1249.4 1285.4
- Garage_Area          1   1265.8 1303.8
- Fireplaces           1   1266.0 1304.0
- First_Flr_SF         1   1313.4 1351.4
- factor(Full_Bath)    4   1426.0 1458.0
```

▼ Code

```
summary(back.model)
```

```
Call:
glm(formula = Bonus ~ factor(House_Style) + Garage_Area + Firepl
    factor(Full_Bath) + factor(Half_Bath) + factor(Central_Air)
    Second_Flr_SF + TotRms_AbvGrd + First_Flr_SF, family = binom
    data = train)

Coefficients:
                                  Estimate Std. Error z valu
(Intercept)                     -1.028e+01  1.541e+00  -6.67
factor(House_Style)One_and_Half_Unf -9.208e+00  3.686e+02  -0.02
factor(House_Style)One_Story     2.062e+00  4.945e-01   4.17
factor(House_Style)SFoyer        1.464e+00  7.213e-01   2.03
factor(House_Style)SLvl          9.390e-01  4.891e-01   1.92
factor(House_Style)Two_and_Half_Fin  1.085e+00  6.908e+00   0.15
factor(House_Style)Two_and_Half_Unf  8.376e-01  8.904e-01   0.94
factor(House_Style)Two_Story     1.010e+00  3.498e-01   2.88
Garage_Area                      3.499e-03  5.210e-04   6.71
Fireplaces                       8.965e-01  1.279e-01   7.01
factor(Full_Bath)1              -6.540e-01  1.330e+00  -0.49
factor(Full_Bath)2               1.930e+00  1.347e+00   1.43
factor(Full_Bath)3               4.355e+00  1.618e+00   2.69
factor(Full_Bath)4              -1.073e+00  2.436e+00  -0.44
factor(Half_Bath)1               1.228e+00  2.215e-01   5.54
factor(Half_Bath)2              -6.069e-02  8.103e-01  -0.07
factor(Central_Air)Y             1.590e+00  5.909e-01   2.69
Second_Flr_SF                    3.466e-03  6.632e-04   5.22
TotRms_AbvGrd                   -4.339e-01  8.142e-02  -5.32
First_Flr_SF                     4.011e-03  4.351e-04   9.22
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2775.8  on 2050  degrees of freedom
Residual deviance: 1214.9  on 2031  degrees of freedom
AIC: 1254.9

Number of Fisher Scoring iterations: 14
```

The model building summary is shown to have removed variables based on lowest AIC. This may be different across different stepwise techniques.

Forward with Interactions

Here we will work through forward selection. In forward selection, the initial model is empty (contains no variables, only the intercept). Each variable is then tried to see if it is significant based on AIC/BIC or p-value with a specified significance level. The most significant variable (or the one that reduces the AIC/BIC the most) is then added to the model. Then the remaining variables are again tested and the next most impactful variable (p-value or AIC/BIC depending on the approach) is added to the model. This process repeats until either no more significant variables are available to add to the model based on AIC/BIC or p-value (depending on the approach). This approach is the same as stepwise selection without the additional check at each step for possible removal.

Forward selection is the least used technique because stepwise selection does the same as forward selection with the added benefit of dropping insignificant variables. The main use for forward selection is to test higher order terms and interactions in models.

Let's look at this approach using all of our softwares!

R Python SAS

In R we must specify two models to step between. We then use the `step` function where we start at the empty model. The `scope` option tells R the lower (`lower =`) and upper (`upper =`) model to step between. The `direction = "forward"` option tells R to use stepwise selection.

Here we are testing some two-way interactions. This can be done in R by creating two models - the main effects model and the two-way interaction model. Here we set the starting point for forward selection at the main effects model and step up to the interaction model.

▼ Code

```
main.model <- glm(Bonus ~ Gr_Liv_Area + factor(House_Style) + G
                          factor(Fireplaces) + factor(Full_Bath
                          factor(Central_Air) + Second_Flr_SF +
                  data = train, family = binomial(link = "logit

int.model <- glm(Bonus ~ Gr_Liv_Area + factor(House_Style) + Ga
                         Fireplaces + factor(Full_Bath) + facto
                         factor(Central_Air) + Second_Flr_SF + 
                         Gr_Liv_Area*factor(House_Style) + TotR
                         Gr_Liv_Area*factor(Fireplaces),
                 data = train, family = binomial(link = "logit

for.model <- step(main.model,
                  scope = list(lower=formula(main.model),
                               upper=formula(int.model)),
                  direction = "forward")
```

```
Start:  AIC=1259.21
Bonus ~ Gr_Liv_Area + factor(House_Style) + Garage_Area + factor
    factor(Full_Bath) + factor(Half_Bath) + Lot_Area + factor(Ce
    Second_Flr_SF + TotRms_AbvGrd + First_Flr_SF

                                    Df Deviance    AIC
+ Gr_Liv_Area:factor(Fireplaces)     3   1159.0 1215.0
+ factor(House_Style):TotRms_AbvGrd  7   1180.8 1244.8
<none>                                   1209.2 1259.2
+ Gr_Liv_Area:factor(House_Style)    7   1202.5 1266.5

Step:  AIC=1215.05
Bonus ~ Gr_Liv_Area + factor(House_Style) + Garage_Area + factor
    factor(Full_Bath) + factor(Half_Bath) + Lot_Area + factor(Ce
    Second_Flr_SF + TotRms_AbvGrd + First_Flr_SF + Gr_Liv_Area:f

                                    Df Deviance    AIC
+ factor(House_Style):TotRms_AbvGrd  7   1125.4 1195.4
<none>                                   1159.0 1215.0
+ Gr_Liv_Area:factor(House_Style)    7   1154.9 1224.9

Step:  AIC=1195.42
Bonus ~ Gr_Liv_Area + factor(House_Style) + Garage_Area + factor
    factor(Full_Bath) + factor(Half_Bath) + Lot_Area + factor(Ce
    Second_Flr_SF + TotRms_AbvGrd + First_Flr_SF + Gr_Liv_Area:f
    factor(House_Style):TotRms_AbvGrd

                                  Df Deviance    AIC
+ Gr_Liv_Area:factor(House_Style)  7   1107.3 1191.3
<none>                                 1125.4 1195.4

Step:  AIC=1191.27
Bonus ~ Gr_Liv_Area + factor(House_Style) + Garage_Area + factor
    factor(Full_Bath) + factor(Half_Bath) + Lot_Area + factor(Ce
    Second_Flr_SF + TotRms_AbvGrd + First_Flr_SF + Gr_Liv_Area:f
    factor(House_Style):TotRms_AbvGrd + Gr_Liv_Area:factor(House

       Df Deviance    AIC
<none>      1107.3 1191.3
```

▼ Code

```
summary(for.model)
```

```
Call:
glm(formula = Bonus ~ Gr_Liv_Area + factor(House_Style) + Garage
    factor(Fireplaces) + factor(Full_Bath) + factor(Half_Bath) +
    Lot_Area + factor(Central_Air) + Second_Flr_SF + TotRms_AbvG
    First_Flr_SF + Gr_Liv_Area:factor(Fireplaces) + factor(House
    Gr_Liv_Area:factor(House_Style), family = binomial(link = "l
    data = train)

Coefficients: (1 not defined because of singularities)
                                                 Estimate Std
(Intercept)                                    -1.065e+01  2.
Gr_Liv_Area                                    -2.722e-03  4.
factor(House_Style)One_and_Half_Unf            -9.155e+00  8.
factor(House_Style)One_Story                    5.159e+00  2.
factor(House_Style)SFoyer                       7.133e-01  4.
factor(House_Style)SLvl                         7.311e+00  3.
factor(House_Style)Two_and_Half_Fin            -4.598e+01  7.
factor(House_Style)Two_and_Half_Unf            -6.236e+00  9.
factor(House_Style)Two_Story                    1.964e+00  2.
Garage_Area                                     3.792e-03  5.
factor(Fireplaces)1                            -3.668e-01  8.
factor(Fireplaces)2                            -4.988e+00  2.
factor(Fireplaces)3                             1.112e+01  2.
factor(Fireplaces)4                             1.533e+01  2.
factor(Full_Bath)1                             -2.696e+00  1.
factor(Full_Bath)2                              8.578e-02  1.
factor(Full_Bath)3                              2.610e+00  1.
factor(Full_Bath)4                             -3.617e+00  4.
factor(Half_Bath)1                              1.336e+00  2.
factor(Half_Bath)2                             -1.039e+00  9.
Lot_Area                                        1.607e-05  1.
factor(Central_Air)Y                            1.974e+00  6.
Second_Flr_SF                                   8.073e-03  4.
TotRms_AbvGrd                                  -7.848e-01  3.
First_Flr_SF                                    9.010e-03  4.
Gr_Liv_Area:factor(Fireplaces)1                 8.676e-04  5.
Gr_Liv_Area:factor(Fireplaces)2                 4.200e-03  1.
Gr_Liv_Area:factor(Fireplaces)3                -4.849e-03  1.
Gr_Liv_Area:factor(Fireplaces)4                        NA
factor(House_Style)One_and_Half_Unf:TotRms_AbvGrd  1.290e+00  1.
factor(House_Style)One_Story:TotRms_AbvGrd     -2.019e-01  4.
factor(House_Style)SFoyer:TotRms_AbvGrd         2.357e+00  8.
factor(House_Style)SLvl:TotRms_AbvGrd          -5.524e-01  7.
factor(House_Style)Two_and_Half_Fin:TotRms_AbvGrd  2.603e+00  8.
factor(House_Style)Two_and_Half_Unf:TotRms_AbvGrd  1.204e+00  1.
factor(House_Style)Two_Story:TotRms_AbvGrd      8.543e-01  4.
Gr_Liv_Area:factor(House_Style)One_and_Half_Unf  -5.634e-03  7.
Gr_Liv_Area:factor(House_Style)One_Story       -9.365e-04  1.
Gr_Liv_Area:factor(House_Style)SFoyer          -9.801e-03  3.
Gr_Liv_Area:factor(House_Style)SLvl            -1.487e-03  2.
Gr_Liv_Area:factor(House_Style)Two_and_Half_Fin   6.811e-03  1.
Gr_Liv_Area:factor(House_Style)Two_and_Half_Unf  -1.516e-03  5.
Gr_Liv_Area:factor(House_Style)Two_Story       -3.911e-03  1.
                                               Pr(>|z|)
(Intercept)                                    7.07e-05 ***
Gr_Liv_Area                                     0.55662
factor(House_Style)One_and_Half_Unf             0.99910
factor(House_Style)One_Story                    0.01934 *
factor(House_Style)SFoyer                       0.86645
factor(House_Style)SLvl                         0.02146 *
factor(House_Style)Two_and_Half_Fin             0.99509
factor(House_Style)Two_and_Half_Unf             0.52598
factor(House_Style)Two_Story                    0.37850
Garage_Area                                    5.98e-11 ***
factor(Fireplaces)1                             0.67034
factor(Fireplaces)2                             0.02685 *
factor(Fireplaces)3                            1.22e-05 ***
factor(Fireplaces)4                             0.99490
factor(Full_Bath)1                              0.07370 .
factor(Full_Bath)2                              0.95492
factor(Full_Bath)3                              0.14052
factor(Full_Bath)4                              0.43468
factor(Half_Bath)1                             5.86e-08 ***
factor(Half_Bath)2                              0.27962
Lot_Area                                        0.36696
factor(Central_Air)Y                            0.00297 **
Second_Flr_SF                                   0.07948 .
TotRms_AbvGrd                                   0.04145 *
First_Flr_SF                                    0.05001 .
Gr_Liv_Area:factor(Fireplaces)1                 0.12202
Gr_Liv_Area:factor(Fireplaces)2                 0.00313 **
Gr_Liv_Area:factor(Fireplaces)3                4.61e-06 ***
```

```
Gr_Liv_Area:factor(Fireplaces)4                                  NA
factor(House_Style)One_and_Half_Unf:TotRms_AbvGrd  0.99921
factor(House_Style)One_Story:TotRms_AbvGrd         0.61913
factor(House_Style)SFoyer:TotRms_AbvGrd            0.00608 **
factor(House_Style)SLvl:TotRms_AbvGrd              0.44621
factor(House_Style)Two_and_Half_Fin:TotRms_AbvGrd  0.99752
factor(House_Style)Two_and_Half_Unf:TotRms_AbvGrd  0.23258
factor(House_Style)Two_Story:TotRms_AbvGrd         0.03573 *
Gr_Liv_Area:factor(House_Style)One_and_Half_Unf    0.99937
Gr_Liv_Area:factor(House_Style)One_Story           0.62877
Gr_Liv_Area:factor(House_Style)SFoyer              0.00140 **
Gr_Liv_Area:factor(House_Style)SLvl                0.55389
Gr_Liv_Area:factor(House_Style)Two_and_Half_Fin    0.99724
Gr_Liv_Area:factor(House_Style)Two_and_Half_Unf    0.77654
Gr_Liv_Area:factor(House_Style)Two_Story           0.03575 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2775.8  on 2050  degrees of freedom
Residual deviance: 1107.3  on 2009  degrees of freedom
AIC: 1191.3

Number of Fisher Scoring iterations: 15
```

Notice that the modeling summary contains the addition of a couple of these interactions.

Diagnostic Plots

Linear regression models contain residuals with properties that are very useful for model diagnostics. However, what is a residual in a logistic regression model? Since we don't have actual probabilities to compare our predicted probabilities against, residuals are not as clearly defined. Instead we have pseudo "residuals" in logistic regression that we can explore further. Two examples of this are **deviance residuals** and **Pearson residuals**.

Deviance is a measure of how far a fitted model is from the fully saturated model. The fully saturated model is a model that predicts our data perfectly by essentially overfitting to it - a variable for each unique combination of inputs. This makes this model impractical for use, but good for comparison. The deviance is essentially our "error" from this "perfect" model. Logistic regression minimizes the sum of the squared deviances. Deviance residuals tell us how much each observation reduces the deviance.

Pearson residuals on the other hand tell us how much each observation changes the Pearson Chi-squared test for the overall model.

Other forms of measuring an observation's influence on the logistic regression model are **DFBetas** and **Cook's D**. Similar to their interpretation in linear regression, these two calculations tell us how each observation changes the estimation of each parameter individually (DFBeta) or how each observation changes the estimation of all the parameters holistically (Cook's D).

Let's see how to get all of these from our softwares!

R
Python
SAS

R has some wonderful diagnostics plots to show us these residuals. R also produces a list of these measures of influence as well as many more with the `influence.measures` function. Below only the first 6 observations are shown using the `head` function, but this is calculated for each of the observations. The 4th plot in the `plot` function on the logistic regression model object is the Cook's D plot as shown below. The `dfbetasPlots` function produces the DFBetas plots, but only one is shown here.

▼ Code

```
logit.model <- glm(Bonus ~ Gr_Liv_Area + factor(House_Style) +
                   Fireplaces + factor(Full_Bath) + Lot_
                   TotRms_AbvGrd + Gr_Liv_Area:Fireplac
               data = train, family = binomial(link = "logi
summary(logit.model)
```

```

Call:
glm(formula = Bonus ~ Gr_Liv_Area + factor(House_Style) + Garage
    Fireplaces + factor(Full_Bath) + Lot_Area + factor(Central_A
    TotRms_AbvGrd + Gr_Liv_Area:Fireplaces, family = binomial(li
    data = train)

Coefficients:
                                     Estimate Std. Error z valu
(Intercept)                        -1.118e+01  1.541e+00  -7.25
Gr_Liv_Area                         4.348e-03  4.804e-04   9.05
factor(House_Style)One_and_Half_Unf -9.494e+00  3.543e+02  -0.02
factor(House_Style)One_Story        2.168e+00  3.242e-01   6.68
factor(House_Style)SFoyer           1.320e+00  6.280e-01   2.10
factor(House_Style)SLvl             1.248e+00  4.421e-01   2.82
factor(House_Style)Two_and_Half_Fin 1.999e-01  3.650e+00   0.05
factor(House_Style)Two_and_Half_Unf 8.763e-01  8.879e-01   0.98
factor(House_Style)Two_Story        1.656e+00  3.135e-01   5.28
Garage_Area                         3.709e-03  5.076e-04   7.30
Fireplaces                          1.817e+00  5.604e-01   3.24
factor(Full_Bath)1                 -4.051e-01  1.198e+00  -0.33
factor(Full_Bath)2                  1.884e+00  1.198e+00   1.57
factor(Full_Bath)3                  3.716e+00  1.510e+00   2.46
factor(Full_Bath)4                 -2.103e+00  2.157e+00  -0.97
Lot_Area                            9.787e-06  1.233e-05   0.79
factor(Central_Air)Y                2.087e+00  5.883e-01   3.54
TotRms_AbvGrd                      -4.835e-01  8.142e-02  -5.93
Gr_Liv_Area:Fireplaces             -5.791e-04  3.546e-04  -1.63
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2775.8  on 2050  degrees of freedom
Residual deviance: 1250.0  on 2032  degrees of freedom
AIC: 1288

Number of Fisher Scoring iterations: 14
```

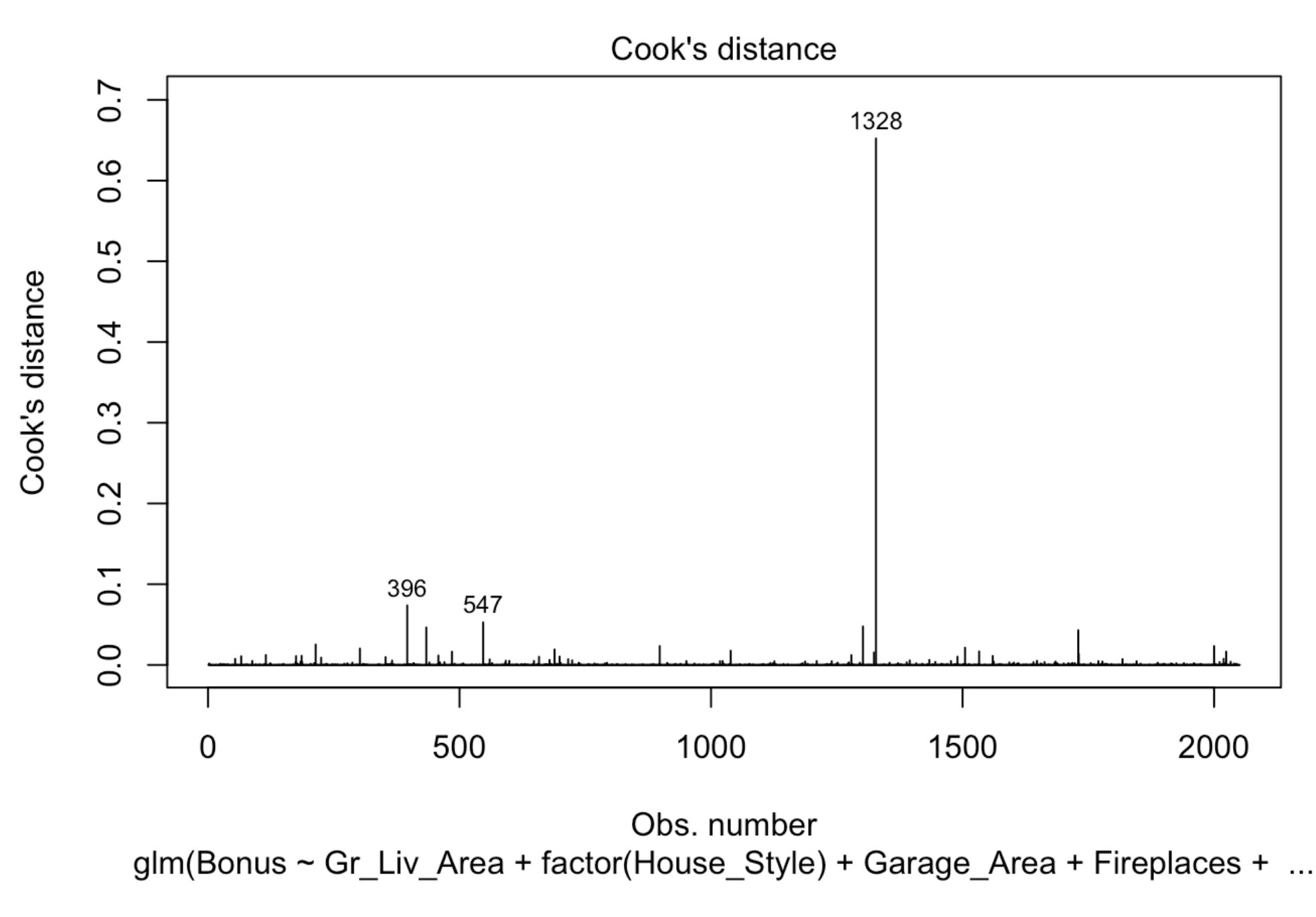
▼ Code

```
library(car)

head(influence.measures(logit.model)$infmat)
```

```
        dfb.1_    dfb.Gr_L_A dfb.f(H_S)O__ dfb.f(H_S)O_S  dfb.f(
1 -0.006680632 -0.0072246691  5.026970e-06   0.002671788 -0.0038
2 -0.023071142 -0.1037195093  3.218790e-06   0.001777788 -0.0036
3  0.041070300 -0.0666518890 -5.618110e-05  -0.043455201 -0.0126
4 -0.003619157 -0.0045229884  4.163032e-06  -0.001883771 -0.0018
5 -0.000551361  0.0004377925  1.408439e-06   0.001199072  0.0006
6 -0.034257763  0.0431126015  2.029811e-05   0.022630715  0.0145
  dfb.f(H_S)SL dfb.f(H_S)T__H_F dfb.f(H_S)T__H_U dfb.f(H_S)T_S
1 -0.0035911372     9.150748e-04     0.0002096029  -0.002443371
2 -0.0128525698    -3.377638e-03    -0.0078565055  -0.039414133
3 -0.0128002414    -6.632733e-04    -0.0006218637   0.001313573
4 -0.0009298446    -1.285294e-04     0.0003646729   0.002090112
5  0.0008280739     9.525692e-05     0.0003348043   0.001098389
6  0.0107742453    -2.817776e-03    -0.0072876282  -0.020112955
       dfb.Frpl   dfb.f(F_B)1  dfb.f(F_B)2  dfb.f(F_B)3   dfb.f(
1  0.0069303163  1.354963e-03 3.559191e-03 0.0045678387  1.35020
2 -0.0054902484  7.121198e-03 4.333448e-03 0.0090488614  1.06392
3 -0.0280704158  3.696606e-03 2.393670e-04 0.0070170929  2.39896
4 -0.0014434218  6.992887e-04 8.272848e-04 0.0019502651 -1.23600
5  0.0004213386 -1.076991e-06 8.822963e-05 0.0001125045 -1.17006
6  0.0386676850  4.075327e-03 1.367289e-03 0.0011850399  1.77490
       dfb.Lt_A      dfb.f(C_      dfb.TR_A   dfb.G_L_A:
1 -6.896558e-04 -7.322808e-04  1.031403e-02 -0.003533782  0.0400
2  3.005167e-02 -9.678781e-03  1.869810e-01  0.018362031 -0.2320
3 -8.319612e-02 -2.455195e-02  1.578453e-02  0.047761640 -0.1756
4 -2.522193e-03  2.814382e-04  6.476448e-03  0.002691227  0.0219
5  9.081321e-05 -3.547551e-05 -6.129572e-05 -0.000324238 -0.0016
6  1.083388e-02  5.567351e-03  2.724948e-02 -0.035826886 -0.1170
        cook.d          hat
1 2.765406e-05 0.0037996009
2 1.650901e-03 0.0137632240
3 1.233413e-03 0.0058384415
4 7.966761e-06 0.0038368812
5 4.200650e-08 0.0003381361
6 2.814123e-04 0.0090119822
```

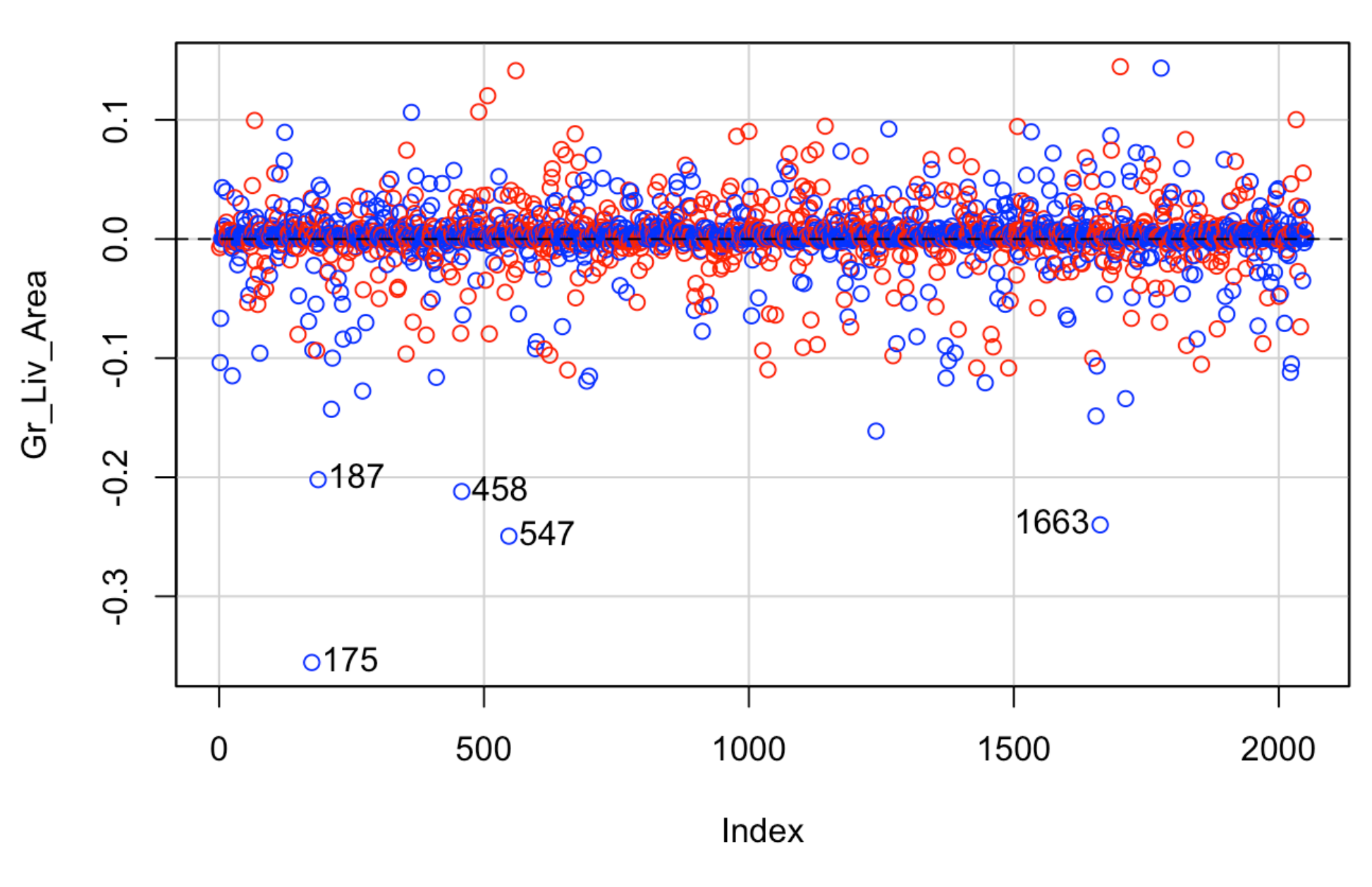
▼ Code

```
plot(logit.model, 4)
```



▼ Code

```
dfbetasPlots(logit.model, terms = "Gr_Liv_Area", id.n = 5,
             col = ifelse(logit.model$y == 1, "red", "blue"))
```



The DFBetas plots show the standardized impact of each observation on the calculation of each of the parameters in the model. The main thing to look for in these plots are points that are far away from the rest of the observations.

These observations are not necessarily bad per se, but have a large influence on the model. These points might need to be investigated further to see if they are actually valid observations.

Calibration Curves

Another evaluation/diagnostic for logistic regression is the **calibration curve**. The calibration curve is a goodness-of-fit measure for logistic regression. Calibration measures how well predicted probabilities agree with actual frequency counts of outcomes (estimated linearly across the data set). These curves can help detect if predictions are consistently too high or low in your model.

If the curve is above the diagonal line, this indicates the model is predicting **lower** probabilities than actually observed. The opposite is true if the curve is below the diagonal line.

This is best used on larger samples since we are calculating the observed proportion of events in the data. In smaller samples this relationship is extrapolated out from the center and may not as accurately reflect the truth.

Let's look at creating these in all of our softwares!

R
Python
SAS

R produces a calibration curve using the `givitiCalibrationBelt` function. The inputs to this function are `o =` and `e =`. These are the observed target and expected target respectively. We place our actual target variable in the `o =` option and the predictions from our logistic regression model in the `e =` option. Since the model is being compared to training data the `devel = internal` option is specified. The `maxDeg =` option sets the maximum degree being tested for the curve.

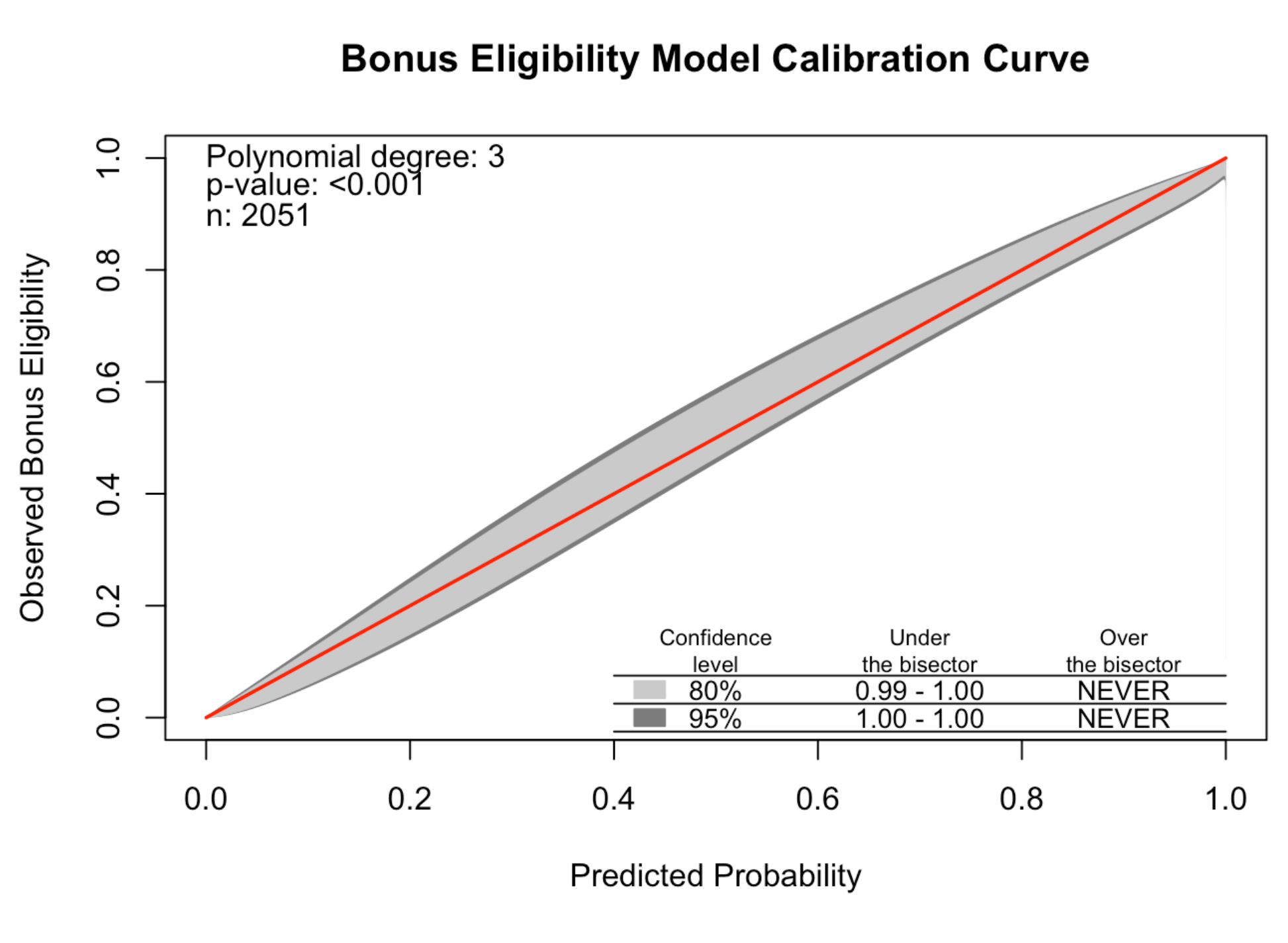
▼ Code

```
logit.model <- glm(Bonus ~ Gr_Liv_Area + factor(House_Style) +
                   Fireplaces + factor(Full_Bath) + Lot_
                   TotRms_AbvGrd + Gr_Liv_Area:Fireplac
               data = train, family = binomial(link = "logi
summary(logit.model)
```

```

Call:
glm(formula = Bonus ~ Gr_Liv_Area + factor(House_Style) + Garage
    Fireplaces + factor(Full_Bath) + Lot_Area + factor(Central_A
    TotRms_AbvGrd + Gr_Liv_Area:Fireplaces, family = binomial(li
    data = train)

Coefficients:
                                     Estimate Std. Error z valu
(Intercept)                        -1.118e+01  1.541e+00  -7.25
Gr_Liv_Area                         4.348e-03  4.804e-04   9.05
factor(House_Style)One_and_Half_Unf -9.494e+00  3.543e+02  -0.02
factor(House_Style)One_Story        2.168e+00  3.242e-01   6.68
factor(House_Style)SFoyer           1.320e+00  6.280e-01   2.10
factor(House_Style)SLvl             1.248e+00  4.421e-01   2.82
factor(House_Style)Two_and_Half_Fin 1.999e-01  3.650e+00   0.05
factor(House_Style)Two_and_Half_Unf 8.763e-01  8.879e-01   0.98
factor(House_Style)Two_Story        1.656e+00  3.135e-01   5.28
Garage_Area                         3.709e-03  5.076e-04   7.30
Fireplaces                          1.817e+00  5.604e-01   3.24
factor(Full_Bath)1                 -4.051e-01  1.198e+00  -0.33
factor(Full_Bath)2                  1.884e+00  1.198e+00   1.57
factor(Full_Bath)3                  3.716e+00  1.510e+00   2.46
factor(Full_Bath)4                 -2.103e+00  2.157e+00  -0.97
Lot_Area                            9.787e-06  1.233e-05   0.79
factor(Central_Air)Y                2.087e+00  5.883e-01   3.54
TotRms_AbvGrd                      -4.835e-01  8.142e-02  -5.93
Gr_Liv_Area:Fireplaces             -5.791e-04  3.546e-04  -1.63
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2775.8  on 2050  degrees of freedom
Residual deviance: 1250.0  on 2032  degrees of freedom
AIC: 1288

Number of Fisher Scoring iterations: 14
```

▼ Code

```
library(givitiR)

cali.curve <- givitiCalibrationBelt(o = train$Bonus,
                                    e = predict(logit.model, ty
                                    devel = "internal",
                                    maxDeg = 5)
plot(cali.curve, main = "Bonus Eligibility Model Calibration Cu
                 xlab = "Predicted Probability",
                 ylab = "Observed Bonus Eligibility")
```



```
$m
[1] 3

$p.value
[1] 0
```

Since the diagonal line is contained in the confidence interval for our calibration curve, we do not notice any significant number of over or under predictions.

← Data Considerations
Model Assessment →